DG INTERNS HUB — TECHNICAL NOTES

TOPIC 09: INTEGRATING SNORT ALERTS INTO A SIEM

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TABLE OF CONTENTS

1. Snort NIDS Architecture & Role
2. Snort Rule Syntax & Structure
3. Syslog & Barnyard2 Log Pipeline
4. Parsing & Field Extraction Engine
5. Practical SOC Scenario: NMAP Scan Alert Triage
6. Summary & Best Practices

1. Snort NIDS & Rule Syntax

Snort inspects network packets in real-time against pre-defined rules to detect network attacks such as port scans, SQL injections, and buffer overflows.

```
# Snort Rule Example: SQL Injection Attempt Detection
alert tcp $EXTERNAL_NET any -> $HTTP_SERVERS $HTTP_PORTS (msg:"MALICIOUS SQLi Attempt
Detected"; flow:to_server,established; content:"UNION"; nocase; content:"SELECT"; nocase;
sid:1000005; rev:1;)
```

2. Log Pipeline Integration Workflow

`Snort Packet Inspection → Writes to /var/log/snort/alert → Universal Forwarder / Syslog Daemon → SIEM Parsing Engine (Extracts src_ip, dst_ip, sid, msg) → Indexer.`

3. Practical SOC Scenario: Correlating NMAP Scan with Host Firewall

Scenario: Reconnaissance Detection to Incident Triage

NIDS Layer: Snort generates Rule SID `122:1` alert: *"NMAP SYN Scan Detected"* from source IP `198.51.100.99`.

SIEM Action: SIEM matches Snort NIDS alert with Windows Firewall log showing subsequent connection attempts on SMB (Port 445).

Analyst Workflow: Confirm active reconnaissance, apply temporary perimeter firewall drop rule on source IP, and initiate endpoint scan on target host.